A-A-53818
18 December 1990

COMMERCIAL ITEM DESCRIPTION

SUCTION APPARATUS, DENTAL

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-S-42002 (DM) which is cancelled.

This commercial item description covers a heavy duty dental suction apparatus used during dental surgical procedures.

Salient Characteristics:

General. Shall be a mobile, electric powered suction apparatus used in the aspiration of blood, saliva and irrigating solutions during dental surgical procedures.

Safety standard. The contractor shall provide proof, to the procurement contracting officer, not less than thirty days prior to delivery, that the items to be delivered under the contract conform to the applicable requirements of Underwriter's Laboratories, Inc. (UL) Standard of Safety 544. A copy of the test report shall be made available to the Government inspection personnel for their use in determining that the units being submitted for acceptance are identical to the unit being submitted for approval. Proof shall consist of one of the following:

UL listing and UL label affixed to the product, provided the contractor agrees to have Underwriter's Laboratories, Inc. forward a copy of the UL report directly to the contracting officer, and the Government inspection personnel, upon request.

A label or equivalent means of identification affixed to the product as evidence that the product design has been examined and tested for design safety compliance by a nationally recognized independent laboratory.

If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with UL standards. The Government reserves the right to require a facilities report and to conduct a facilities survey.

AMSC N/A FSC 6520

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Materials. Materials shall conform to the applicable specifications, standards, and requirements specified. Materials not specifically designated shall be of high commercial quality throughout, and shall be entirely suitable for the intended purpose.

Style and design. The dental suction apparatus shall be a mobile, pedestal type designed for operation using 115 volts, 60 hertz, AC. The suction apparatus shall consist of a motor driven pump, a complete suction or aspirating system, a 32 ounce bottle with cap, an adjusting regulator, one 6 foot tubing and one 15 inch tubing. The entire unit shall be compactly mounted on a sturdy pedestal type stand including a full base and a metal box type top. The unit shall be furnished with the following accessories: Tip handle, Tip No. 2, Tip No. 4, one additional 32 ounce bottle without cap and a package of 3 bacteria filters. The suction apparatus shall weigh 53 pounds $\pm$ 5 pounds.

Replacement parts. Replacement parts for the suction apparatus shall be the following: Tip handle, Tip No. 2 coupland type, Tip No. 4 coupland type, tube and bottle cap assembly and bacteria filters. The style and design of the tips and handle shall be in accordance with (DPSC) Defense Personnel Support Center drawing #1143. The tip handle and tips shall be chrome or nickel plated CDA260 brass. The tips shall be secured to the threaded (distal) end of the handle by means of a knurled tightening sleeve. The tip shall fit smoothly onto the handle and shall not be removeable from the handle without loosening the sleeve. The tube and bottle cap assembly shall consist of one rubber cap, one 6 foot length (minimum) of tubing and one 15 inch long (minimum) tube. The tips and handle shall show no evidence of corrosion when tested in accordance with the boil test of ASTM F1089.

Stand. The top and front panel shall be fabricated from painted 16 gauge corrosion resistant steel. The stand top cover shall be fabricated from 18 gauge corrosion resistant steel. The top (motor housing) of the suction apparatus shall have multiple ventilation louvers. The base of the suction apparatus shall have four casters. Two of the casters shall be of the locking type installed diagonally opposite each other. Each caster shall be single-wheeled, shall have rubber tread, and be oilless. The 32 ounce bottle bracket shall be permanently attached to the right side of the stand top. The bottle bracket shall protect the bottle from possible damage. The style and design of the stand shall be in accordance with Figure 1.

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Motor. The pump drive shall be shaded pole, continuous duty, vented and fan cooled. The motor shall have a built-in overload circuit breaker. The motor shall be at least 1/8 horsepower and rated at 1575 revolutions per minute, 115 volts and 60 cycles. The current required to start the motor shall not exceed 16 amperes. The motor shall operate at a maximum of 4.0 amperes with a maximum temperature rise of 40^0 centigrade. The motor shall be in accordance with the requirements of Federal Specification CC-M-1807.

Cord. An electric cord, 5 foot minimum in length, with a "hospital grade" plug shall be provided to permit connection of the suction unit to a receptacle. The cord shall have a power cable strain relief. An "on-off" power switch shall be mounted in the front panel and shall light when the unit is energized.

Dielectric strength. The suction apparatus shall withstand the dielectric strength test as specified herein.

Pump. The pump shall be a direct drive, single head, diaphragm compressor capable of developing a negative pressure of 22 inches of mercury (minimum). The pump shall have sufficient capacity to evacuate air from a 32 ounce bottle to a vacuum of 22 inches of mercury in 6 seconds. The pump shall be capable of 8 hours of continuous operation at a minimum flow rate of 25 liters per minute of air without requiring servicing. Three filter elements, within the pump head, shall filter the air passing through the inlet and outlet ports. Inlet and outlet valves shall be fabricated from corrosion resistant steel. The pump shall have a muffler on the outlet port which shall assure a quiet operation. The muffler shall be designed to permit easy removal and replacement of the filter element.

Lubrication. The motor and pump bearings shall be permanently lubricated at the factory and shall not require any further lubrication.

Bacteria filter. The pump shall be protected from damage from an overfilled suction bottle and contamination by a hydrophobic, 0.3 micron bacteria filter located between the pump and the vacuum bottle. The filter shall have a barb connection on one end designed to accept 1/4 to 1/2 inch flexible tubing. The filter housing shall be fabricated from acrylic and shall be disposable.

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Vacuum regulator and gauge. A needle type vacuum regulator shall be fabricated from plastic and shall be designed to provide manual adjustment to the pressure. A pressure gauge shall be provided with the unit. The gauge shall have a range of 0 to 30 inches of mercury and have a dual scale providing readings in both inches and millimeters of mercury. The vacuum gage shall be accurate to ± 1 inch of mercury when tested using a mercury monometer.

Bottle assembly. The clear glass collection bottle shall have a minimum capacity of 32 ounces. The bottle shall be properly labeled, including the maximum fill line. The bottle shall be designed to withstand a vacuum pressure of 25 inches of mercury. Inlet and outlet tubes shall be provided for attachment of patient tubing and pump tubing. The tubes shall be fabricated of brass, chromium plated, and/or corrosion resistant steel have a slip type connector and provide a proper seal when installed in the cap. The inlet tube shall extend at least 2-5/8 inches farther into the bottle than the outlet tube, in order to prevent splashing.

Cap. A molded, overlapping, oil-proof, chemical-resistant synthetic rubber cap shall be furnished with the bottle. The cap shall have two holes one for the inlet tube and one for the outlet tube. The cap shall form a seal around the lip as well as the top of the bottle.

Tubing. The tubing shall be fabricated from clear flexible plastic and be equipped with plastic connectors designed to adapt to metal tubes. The tubing shall be clean and disposable. The tubing shall consist of one 6-foot (minimum) piece and one 15 inch (minimum) piece.

Caution and warning notices.

Power supply card. Tag attached to power supply cord shall be permanently and legibly marked as follows:

CAUTION

GROUNDING RELIABILITY CAN ONLY
BE ACHIEVED WHEN THE EQUIPMENT
IS CONNECTED TO AN EQUIVALENT
RECEPTACLE MARKED "HOSPITAL GRADE"

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Servicing notice. A suitable notice shall be provided on the suction apparatus where readily visible during any attempt to perform servicing:

CAUTION

ELECTRIC SHOCK HAZARD
DO NOT REMOVE COVER (OR BACK)
REFER SERVICING TO QUALIFIED
SERVICE PERSONNEL

Flammable anesthetics. The suction apparatus shall be provided with a readily visible warning silkscreened on the stand front as follows:

DANGER

POSSIBLE EXPLOSION HAZARD
IF USED IN PRESENCE OF
FLAMMABLE ANESTHETICS

Identification marking. Each suction apparatus unit shall be permanently and legibly marked with the name or registered trademark of the manufacturer, model designation, serial number, and electrical characteristics, such as voltage, cycle, and amperage.

Service data. The contractor shall furnish, with each suction apparatus unit, two copies each of a technical manual which contains complete instructions for installation, operation and maintenance, and lists of component parts. The manual shall comply with the requirements of Military Specification MIL-M-7298.

Finish replacement parts. The finish of the replacement parts shall be mirror or satin.

Finish suction apparatus. All surfaces shall be clean and free of foreign matter. Finish on edges and surfaces shall be uniform and free from burrs, slivers, sharp edges, rough areas, cracks or other defects.

Workmanship. Items shall be free from defects which detracts from their appearance or impairs their serviceability.

Unit. Each (EA). One complete suction assembly, 2 vacuum bottles, hose, handle, 2 tip, and replacement filters such as required, as specified, constitutes one unit. Each suction unit shall come with detailed instructions for use and maintenance.

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after the delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 1.5 (percent defective) and an inspection level of II, except for dimensional defects which shall have an inspection level of S-1. The unit of product for examination purposes shall be one dental suction apparatus.

Examination. Examination shall be conducted to determine compliance with specification requirements.

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Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, with an inspection level of S-2 with an AQL of 1.0. The unit of product shall be one dental suction apparatus.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Vacuum (suction) bottle. The suction bottle shall be subjected to an internal vacuum of 25 inches of mercury for a period of 1 minute without failure.

Performance. The unit shall be operated at 22 inches of mercury in accordance with manufacturer's instructions for a minimum of eight hours. The minimum time required for suction apparatus to reach 22 inches of mercury shall be noted. Using the regulator, vary the suction from 0 to 22 inches of mercury.

Dielectric strength. A dielectric test in accordance with UL requirements shall be performed after the pump has been warmed by operation.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and markings shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document.

National Stock Number	Item Identification
6520-01-206-0299	Suction Apparatus, Dental, Mobile 115 volts, 60 Hz, Stand Mounted

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order.

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MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS
PHS
FDA-MPQAS

Project No. 6520-2069

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NOTE.

THE SUCTION APPARATUS SHOWN IS TYPICAL IN STYLE AND DESIGN AND IS NOT INTENDED TO RESTRICT OR PRECLUDE THE PURCHASE OF SUCTION APPARATUS WHICH ARE OTHERWISE IN ACCORDANCE WITH THE REQUIREMENTS OF THIS DOCUMENT.

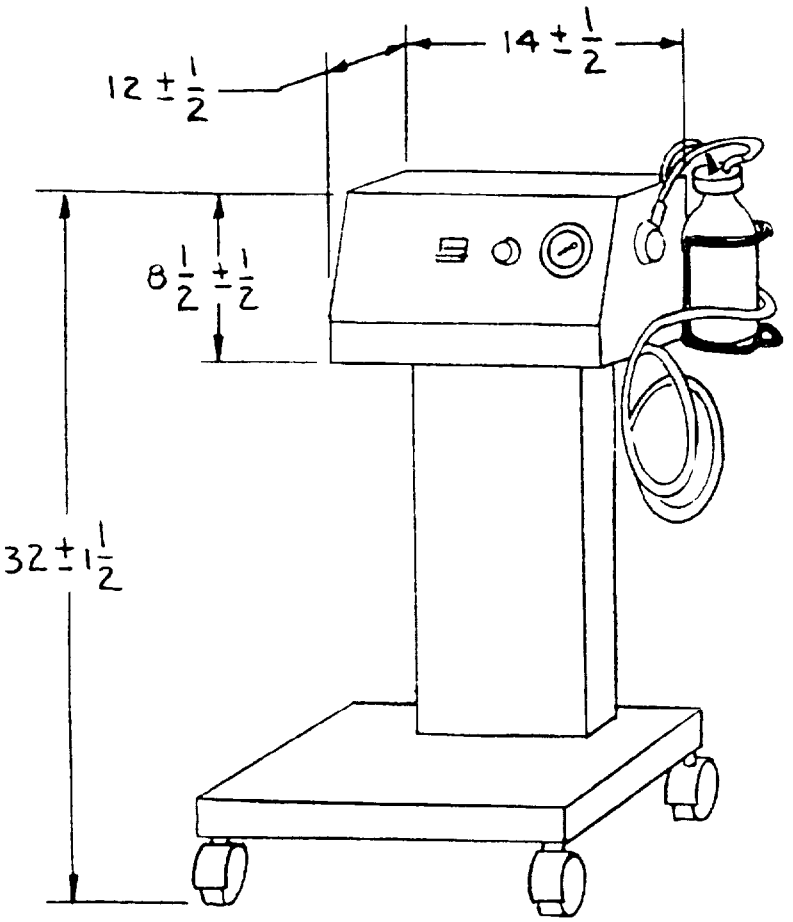


FIGURE 1 SUCTION APPARATUS

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NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-53818	SUCTION APPARATUS, DENTAL

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last First MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street City State ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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